

Photodissociation Dynamics of CH₄ Using PySpawn/OpenMolcas

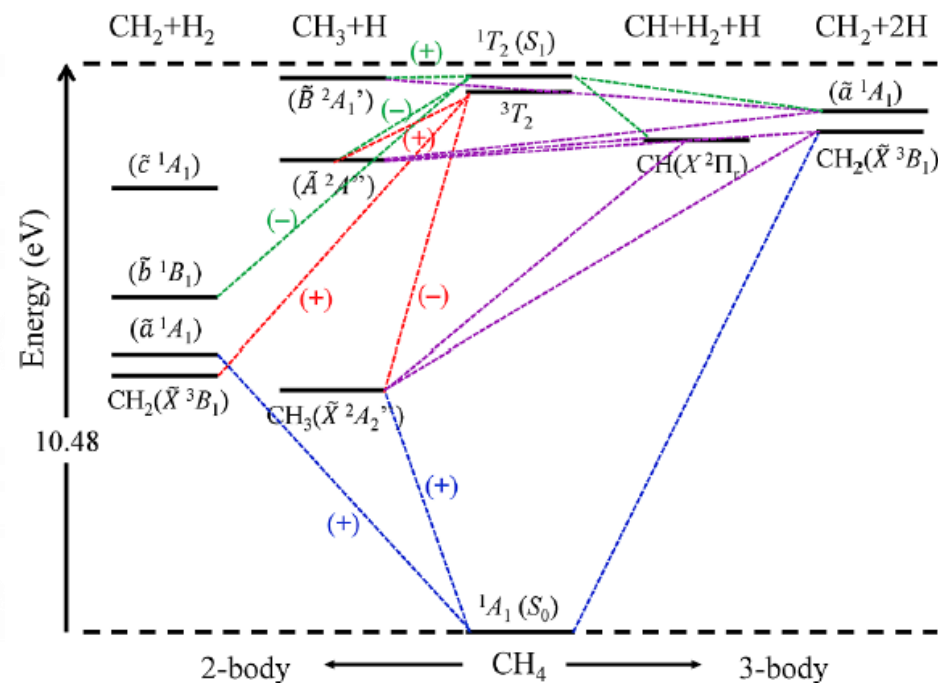
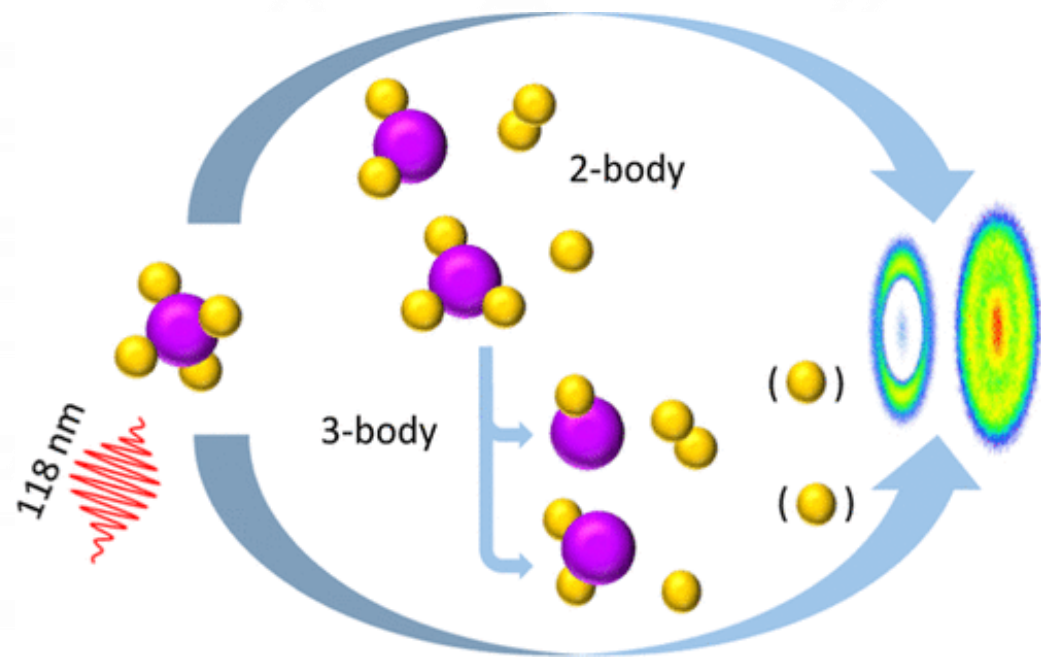
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Reaction pathways of CH₄ photodissociation



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- Reaction pathways of CH₄ photodissociation have been studied.
- The pathways include major 2-body reactions and minor 3-body reactions.

Branching ratio of CH₄ photodissociation



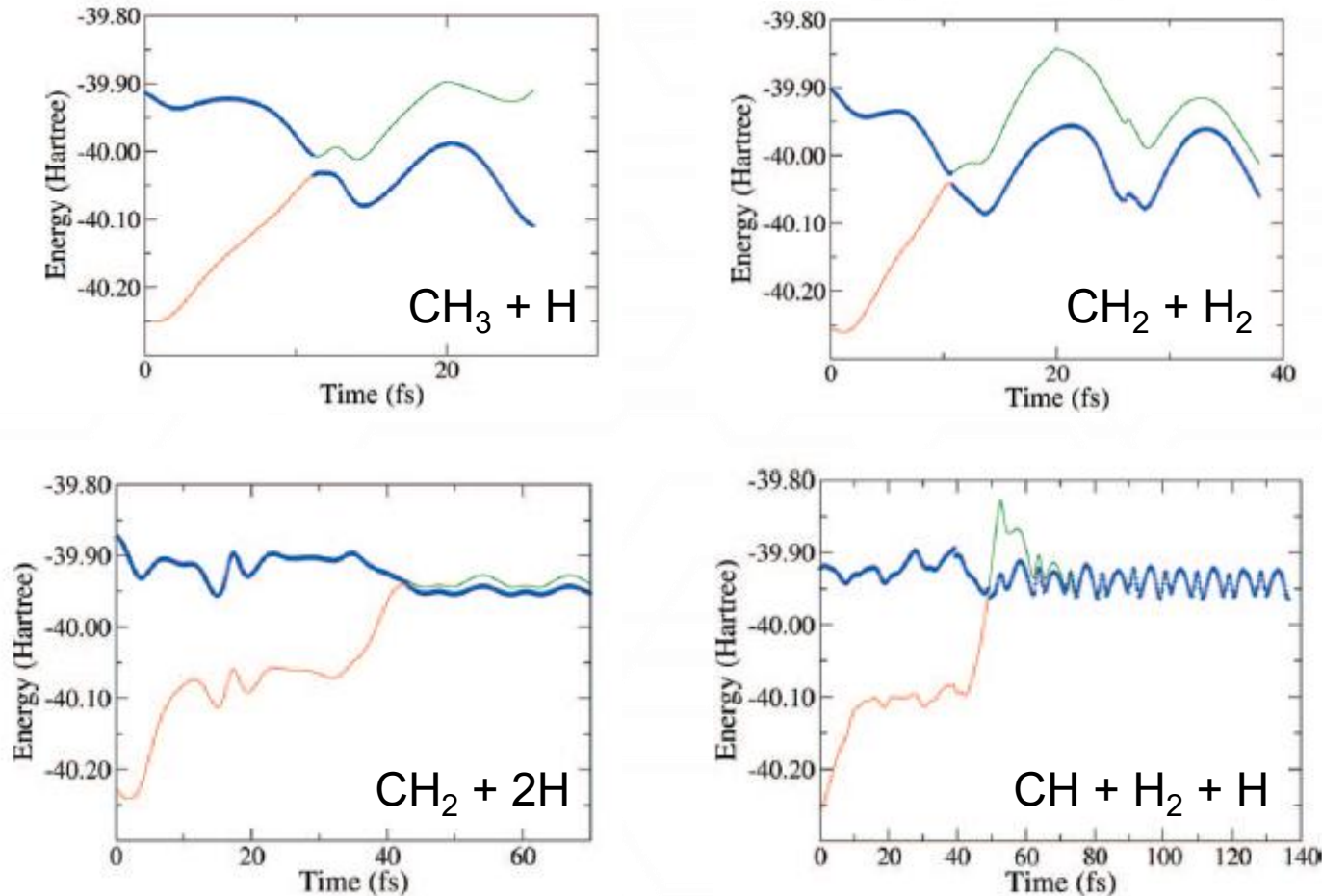
channels		$\lambda = 118 \text{ nm}$		$\lambda = 121.6 \text{ nm}$			
		this work	Ref. 28	Ref. 28	Ref. 19	Ref. 16	Ref. 10
Two body	CH ₃ ($\tilde{X}$)+H(1a)	0.25±0.02	0.26±0.04	0.42±0.05	0.291±0.068	0.39	0.379±0.03
	CH ₂ ($\tilde{X}$)+H ₂ (2a)	} 0.66 ±0.04	≈0	} 0.585 ±0.098	} 0.52	}	0.245±0.027
	CH ₂ ($\tilde{a}$)+H ₂ (2b)		0.17±0.05 ^(a)				
	CH ₂ ($\tilde{b}$)+H ₂ (2c)		≈0				
Three body	CH ₂ ($\tilde{X}$)+2H(3a)	0.042±0.006	0.48±0.06	0.03±0.08	} 0.055 ±0.012	} 0.01	0.058±0.014
	CH ₂ ($\tilde{a}$)+2H(3b)	0.016±0.008		≈0			0.038±0.012
	CH(X)+H+H ₂ (4)	0.029±0.015	0.09	0.07	0.068±0.013	0.08	0.015±0.008

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- The branching ratio of CH₄ photodissociation was reported in previous experimental and computational studies.

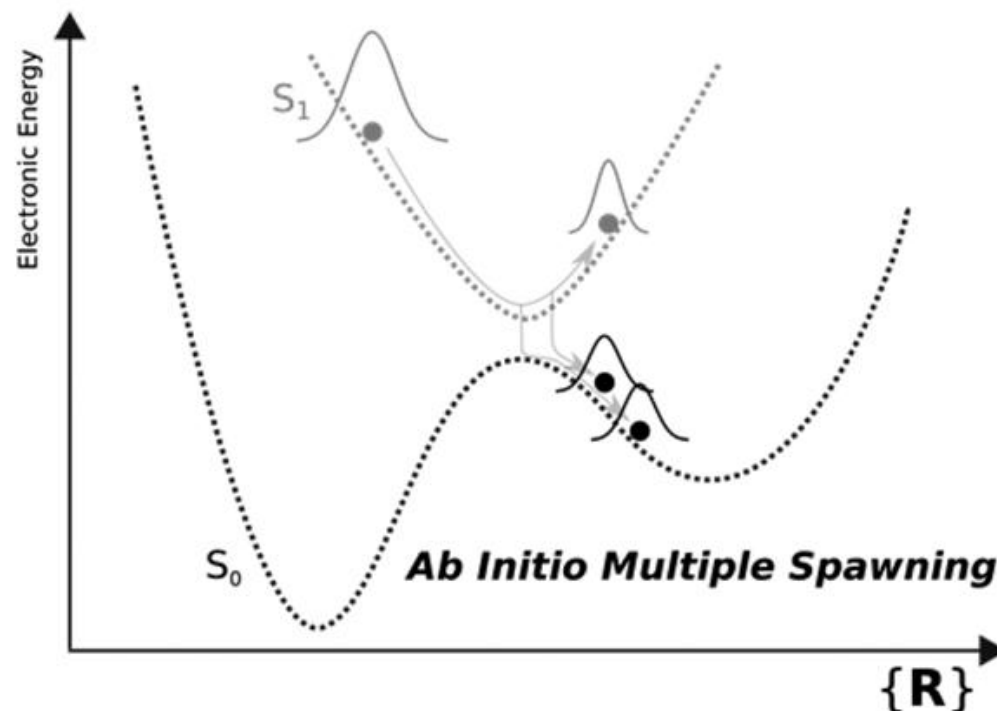
Timescale of CH₄ photodissociation

- Potential energy change in NAMD with TSH (FSSH) method



- CH₄ photodissociation reactions take place within tens of femtoseconds.
- They do not require long simulation times.
- Hence, this system is an efficient testbed for NAMD methods.

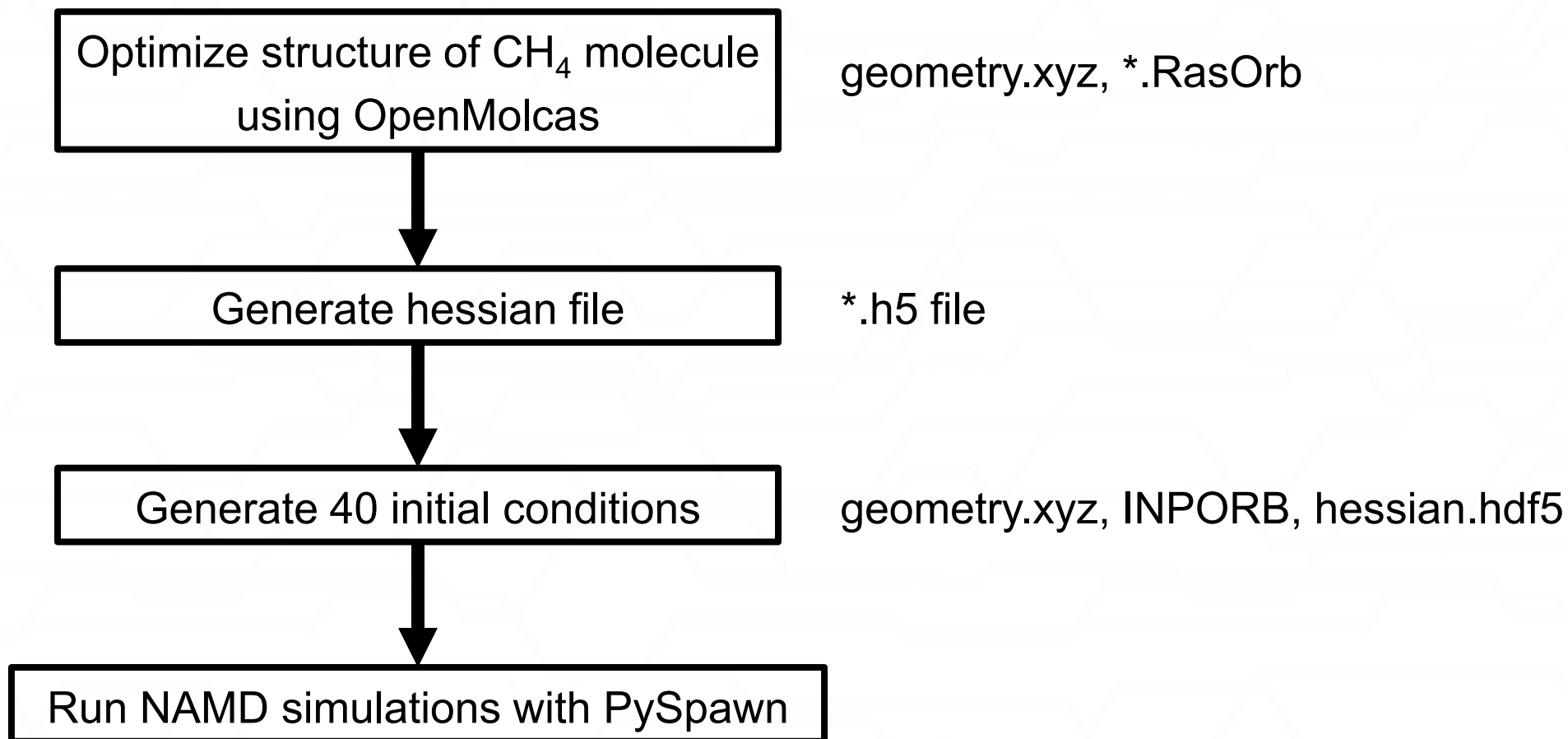
Ab initio multiple spawning (AIMS) method



Chem.Rev. **118**, 3305–3336 (2018)

- Multiple trajectories are spawned during a simulation.
- When applying AIMS to the CH_4 photodissociation case, we can confirm that AIMS generates different reaction pathways within a single simulation.

Workflow



Simulation settings

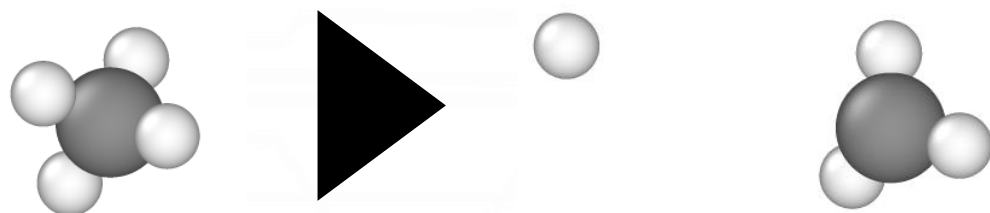


- OpenMolcas
 - ✓ SA(2)-XMS-CASPT2(8,8)
 - ✓ aug-cc-pVDZ basis set
 - ✓ No triplet states included
- PySpawn
 - ✓ Time step (ts): 0.1 fs
 - ✓ Total time (tfinal): 50 fs
 - ✓ Initial state (istate): 1 (excited state)
 - ✓ Coupling threshold (spawnthresh): $(0.5 * \text{np.pi}) / \text{ts} / 40.0$
 - ✓ 32 initial conditions generated

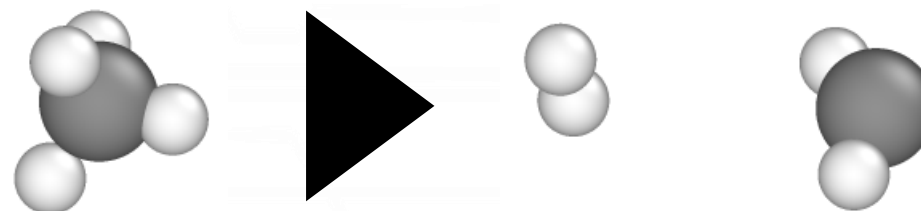
Observed reactions



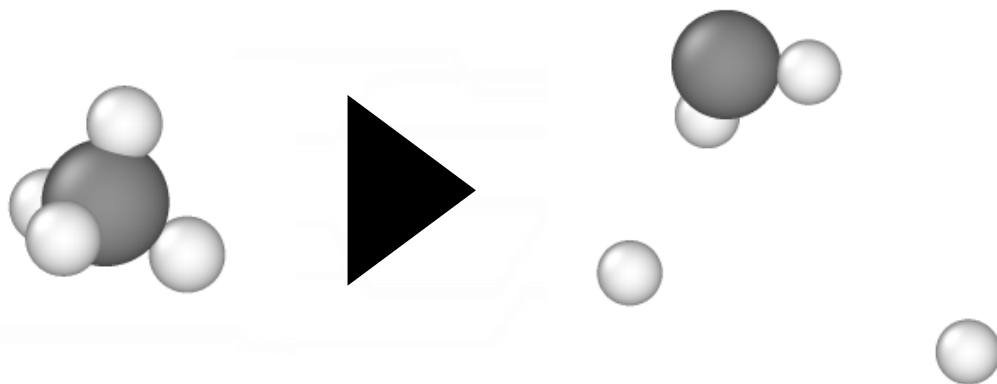
- $\text{CH}_3 + \text{H}$ (initial condition #25)



- $\text{CH}_2 + \text{H}_2$ (initial condition #3)



- $\text{CH}_2 + 2\text{H}$ (initial condition #8)



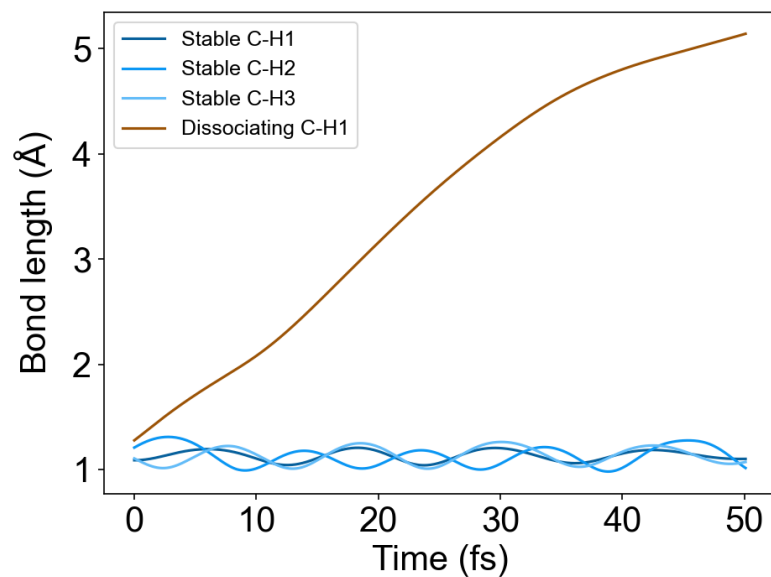
- $\text{CH} + \text{H}_2 + \text{H}$

- ✓ Not observed
- ✓ Simulation time and number of initial conditions are not sufficient

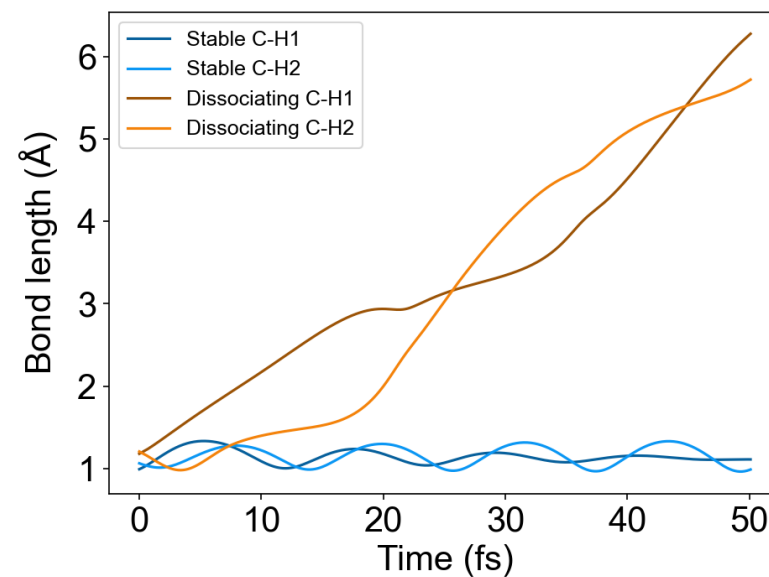
Change in C-H bond lengths



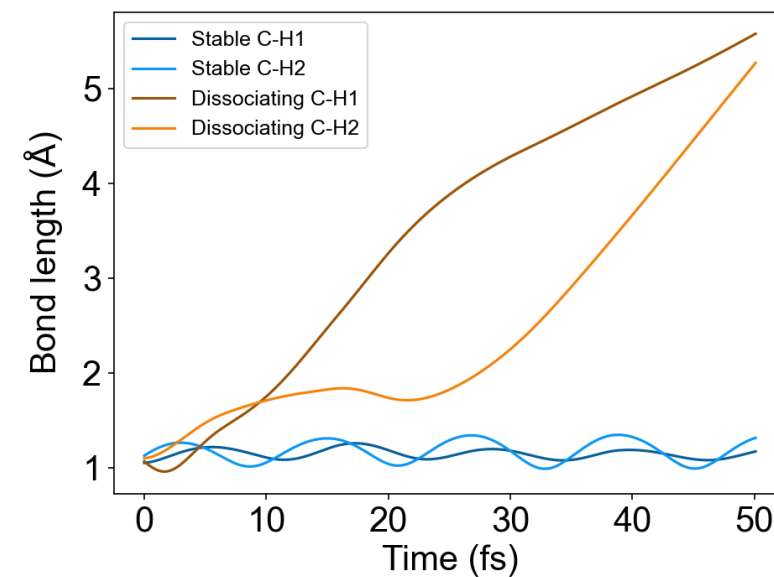
- $\text{CH}_3 + \text{H}$ (initial condition #25)



- $\text{CH}_2 + \text{H}_2$ (initial condition #3)



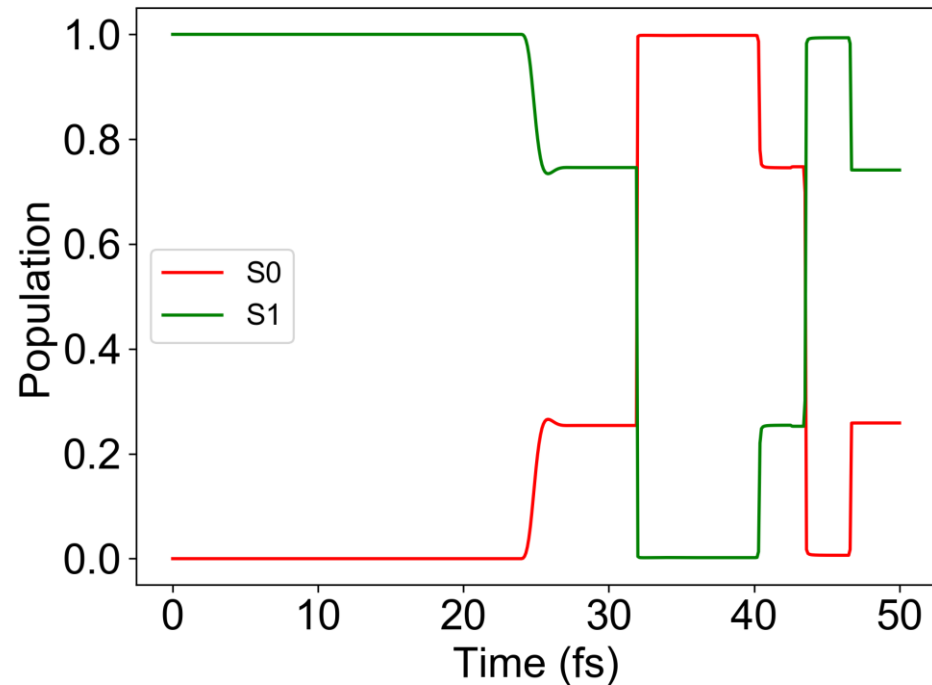
- $\text{CH}_2 + 2\text{H}$ (initial condition #8)



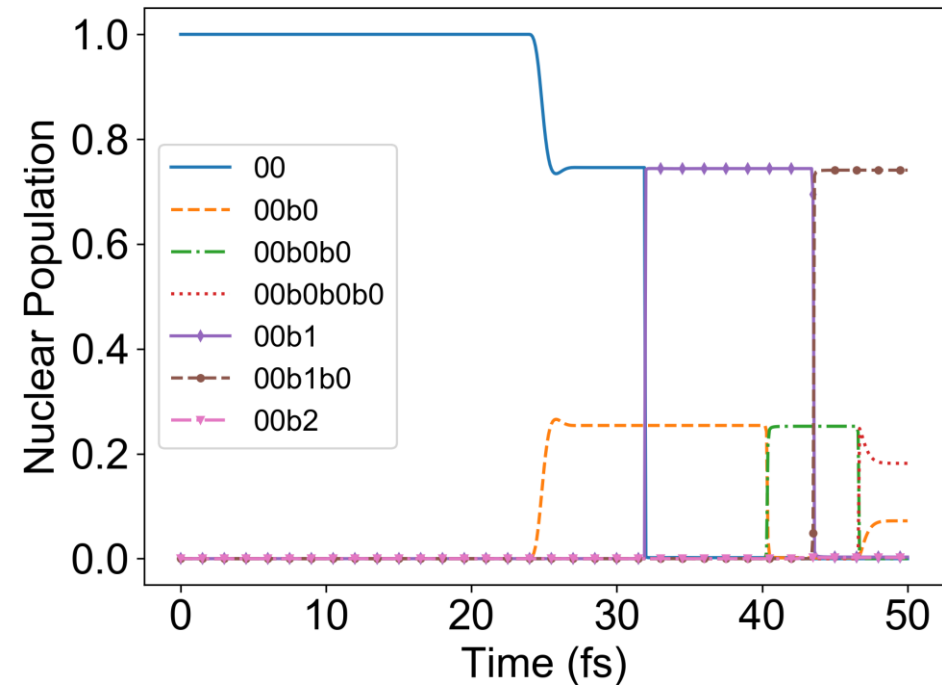
Simulation results of initial condition #32



- Electronic population



- Nuclear population

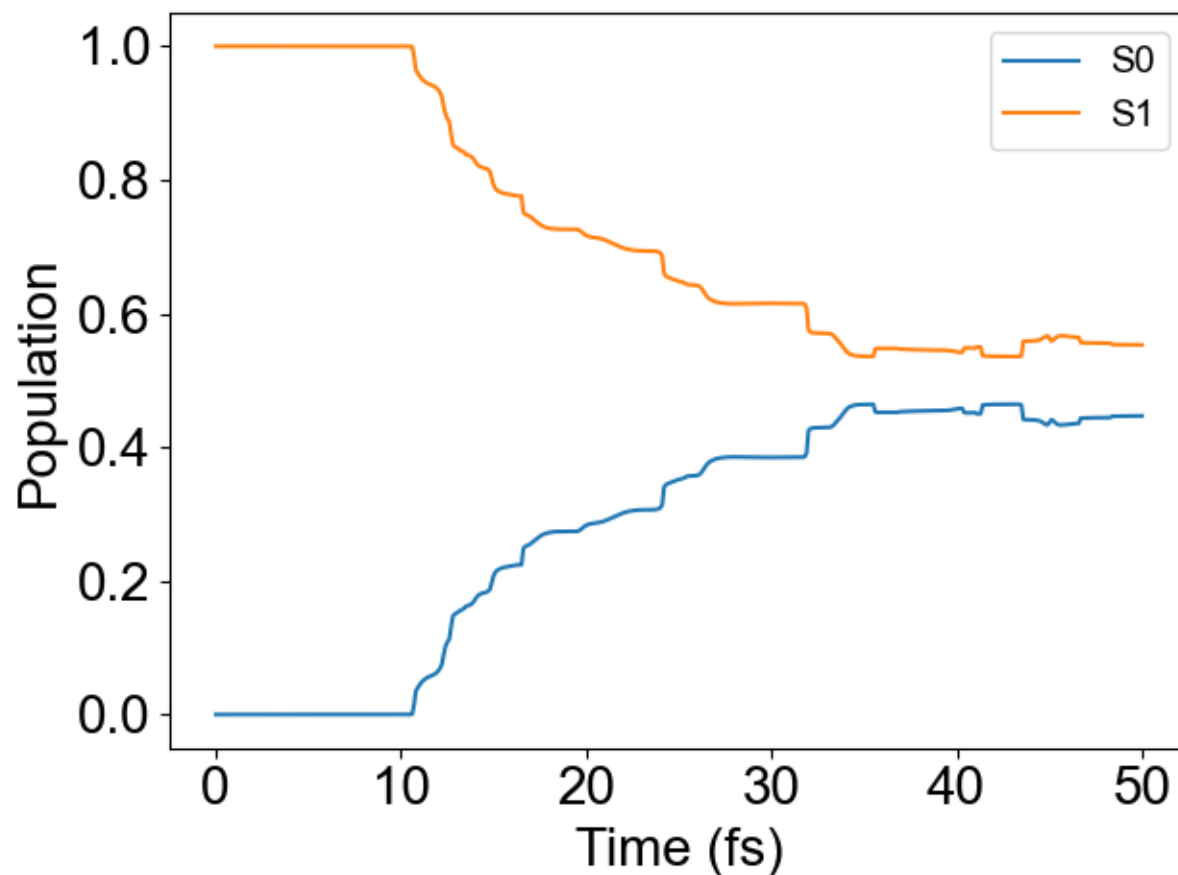


- 00: $\text{CH}_2 + 2\text{H}$
- 00b0: $\text{CH}_2 + 2\text{H}$
- 00b0b0: $\text{CH}_2 + \text{H}_2$
- 00b0b0b0: $\text{CH}_2 + \text{H}_2$
- 00b1: $\text{CH}_2 + 2\text{H}$
- 00b1b0: $\text{CH}_2 + 2\text{H}$
- 00b2: $\text{CH}_2 + 2\text{H}$

Electronic population evolution



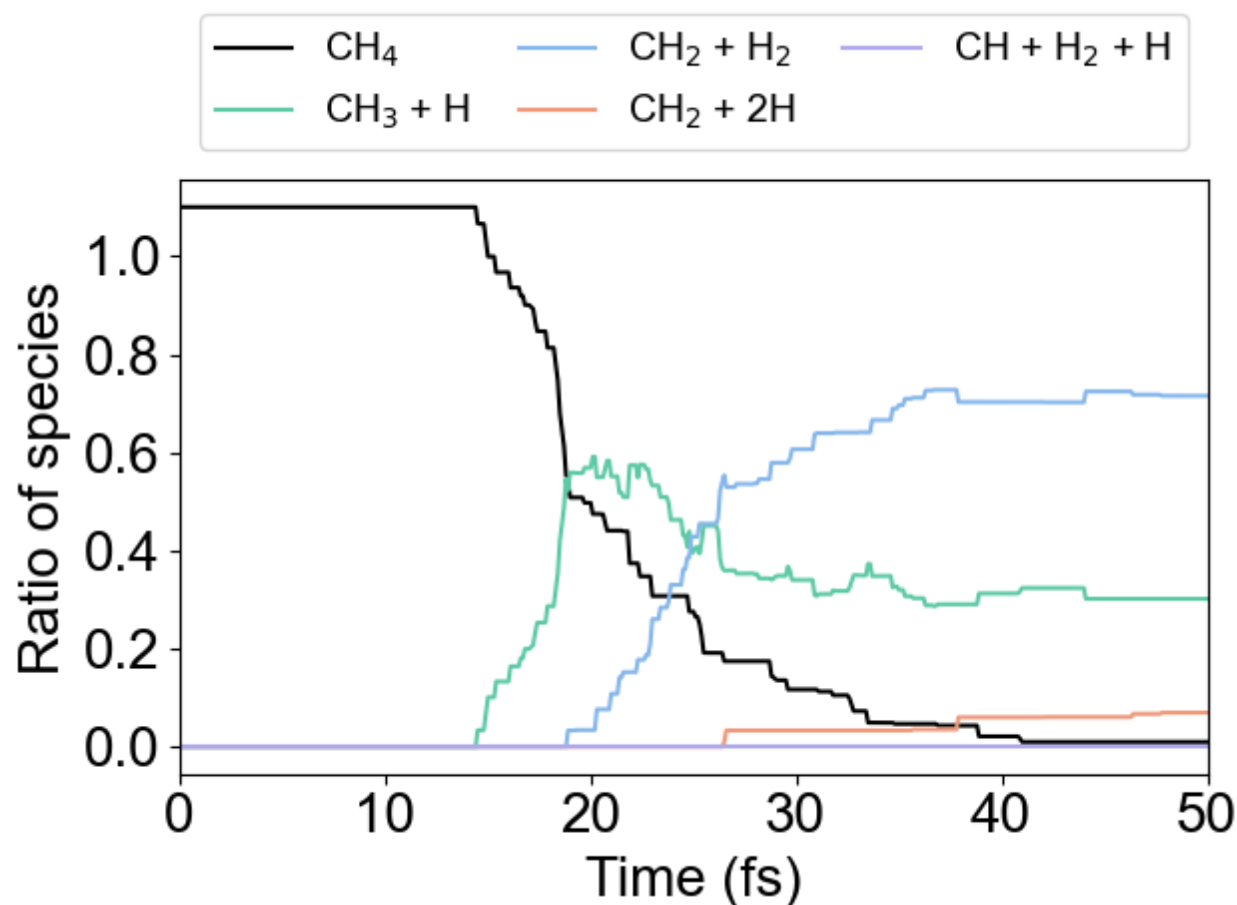
- The electronic population is averaged over all 32 simulations, showing the overall population transfer between electronic states as a function of time.



Branching ratio of different pathways



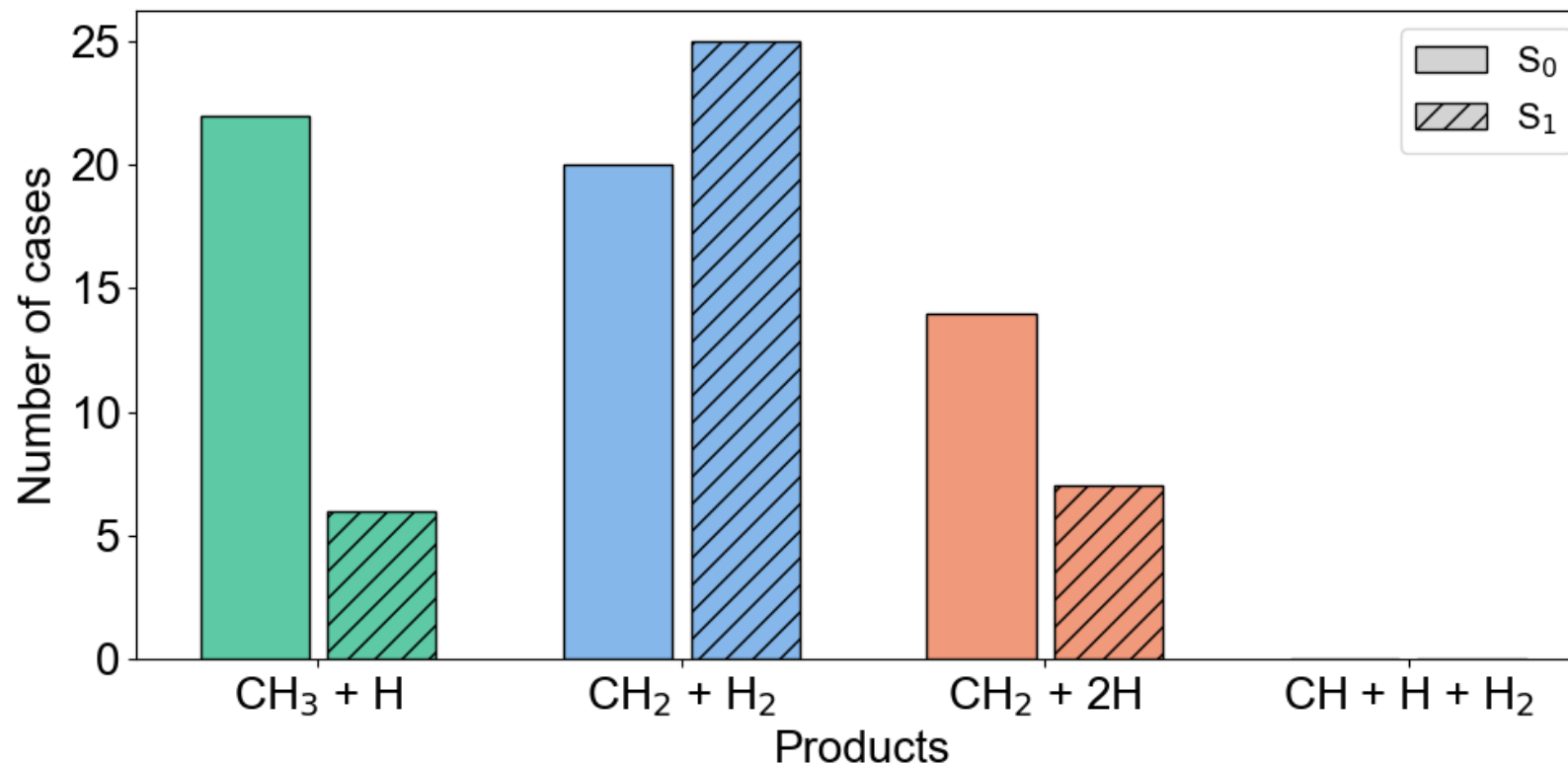
- The ratio of species was calculated by considering the nuclear population of each trajectory across the 32 simulations.



	AIMS	Ref. 1
CH ₃ + H	30%	25%
CH ₂ + H ₂	72%	70%
CH ₂ + 2H	7%	6%
CH + H ₂ + H	0%	3%

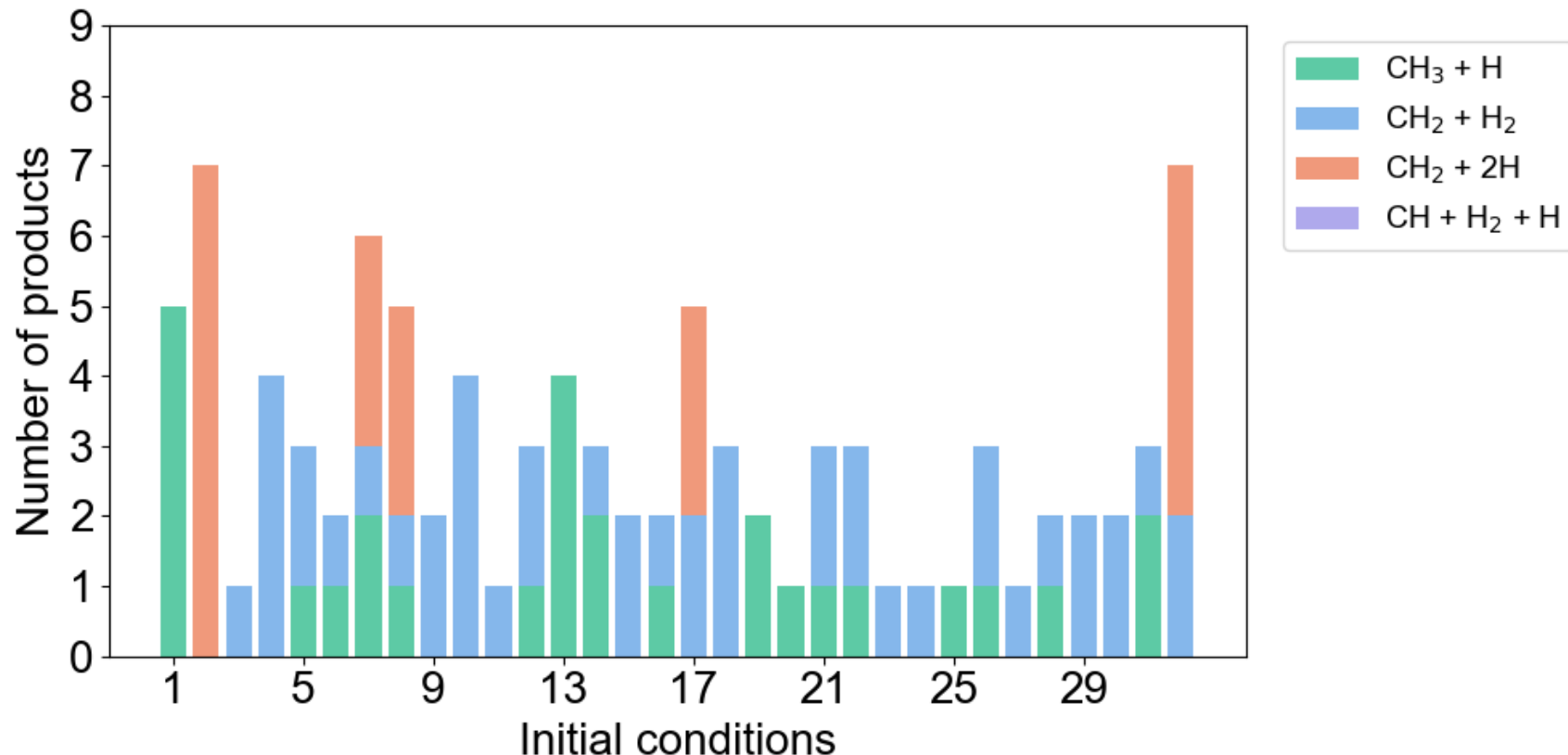
[1] J. Phys. Chem. A **128**, 5273–5284 (2024)

Electronic states of products



- Majority of $\text{CH}_3 + \text{H}$ was produced on ground state (internal conversion $\rightarrow$ dissociation).
- $\text{CH}_2 + \text{H}_2$ was produced on both ground and excited state at similar ratio.
- These are consistent previous study (J. Chem. Phys. **131**, 224320 (2009)).

Number of observed products in each simulation



- In a single simulation, up to three different dissociation pathways were observed.
- In the point of view of sampling geometries for training machine learning potential for NAMD, AIMS method can be efficient way compared to TSH methods.

Summary



- The AIMS method, using PySpawn/OpenMolcas, was applied to NAMD simulations of the CH_4 photodissociation reaction.
- All 32 simulations showed branching ratios and electronic states of products similar to those from previous experimental and computational (TSH method) studies.
- Within a single simulation, multiple trajectories following different reaction pathways were observed.
- Hence, AIMS can be used as an efficient method for collecting training geometries for machine learning models.

